

5 a DERQ and CFSP [B1]

- bi As current flows through the slice in the presence of the magnetic field, it will experience a magnetic force to the left, by Fleming Left Hand rule. Hence a potential difference is generated across DERQ and CFSP.

When the current is undeflected at equilibrium, force on charge due to magnetic field equals force due to electric field [B1]

$$Bqv = Eq \quad [B1]$$

$$E = B v$$

Electric field across faces DERQ and CFSP is constant

$$E = V_H / d, \text{ where } d \text{ is the distance between faces DERQ and CFSP} \quad [A1]$$

$$V_H = Bvd$$

- bii since $I = nAqv$ where [M1]

A is the cross-sectional area through which current flows, $A = dt$, and t is the distance between faces CDEF and PQRS.

$$v = \frac{I}{n(dt)q} \quad [C1]$$

$$V_H = (Bd)\left(\frac{I}{n(dt)q}\right) = \frac{BI}{ntq} \quad [A0]$$

- biii For copper, n is very large hence V_H is very small, making it difficult to measure. [B1]

6 ai The *magnetic flux density at a point* is defined as the force acting per unit current per **11**